Chemistry 3A – Fall 2026 **Version A**

Sections 25810-25811

Midterm Examination #1 1 September 2026

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. What is the maximum number of electrons that can occupy a **single orbital**?
   a) 1 b) 2 c) 4 d) 8

2. Elements in Group 2 (IIA) Main Group elements of the Periodic Table are known as:
   a) alkali metals b) noble gases c) halogens d) alkaline earth metals

3. Which statement is part of Dalton's atomic theory?
   a) All atoms of a given element are identical (have the same identity).
   b) Compounds are formed by a combination of atoms of different elements.
   c) Chemical reactions involve the rearrangement of atoms of different elements
   d) choices (a), (b) and (c) are all statements of Dalton's atomic theory

4. Katherine recorded the mass of chemical product to be 0.00375 grams. She wrote this in scientific notation. What did she write?
   a) 3.75×10^{-5} g b) 3.75×10^{-3} g c) 3.750×10^{-3} g d) 3.75×10^{3} g

5. Which of these in chemistry states that each electron in an atom can have only one set of four quantum numbers unique to the electron?
   a) Hund's Rule b) Pauli Exclusion Principle c) Aufbau Principle d) Dalton's Law

6. A sample with mass 10.0000 g was found to have a density of 2.5 g/mL. What volume did it have, using the correct significant digits/figures?
   a) 25 mL b) 0.25 mL c) 4.0000 mL d) 4.0 mL

7. If a substance has a temperature of 25° C, what is its temperature in Kelvin?
   a) 0 K b) 100 K c) 273 K d) 298 K

8. If someone takes a thermometer outside and says "It's 41°C now", what is that best described as in the scientific method?
   a) observation b) hypothesis c) theory d) experiment

9. The speed of an electromagnetic wave (radiation) is 3.00×10^{8} m/s and its frequency is 5.56×10^{14} s^{-1} (or 5.56×10^{14} Hz). Which value best indicates the wavelength (1 nm = 10^{-9} m)?
   a) 5.40×10^{-7} nm b) 1.85×10^{6} nm c) 540 nm d) 1.67×10^{23} nm

10. Perform the calculation and apply decimal-place rules: 0.0031 + 1.25 – 0.14
    a) 1.113 b) 1.11 c) 1.1 d) 1.1131

11. Which of the following elements is a **non-metal**?
    a) sodium (Na) b) copper (Cu) c) germanium (Ge) d) fluorine (F)

12. In Bohr's model, electrons moving from *higher* to *lower* energy levels would typically:
    a) ***emit*** a photon b) ***absorb*** a photon c) *create* a new **isotope** d) become cations

13. An atom becomes a **cation**. What is TRUE of that atom?
    a) It will have a **positive** charge b) it lost a proton from its nucleus
    c) its mass number (***A***) changed d) choices (a), (b), and (c) are all true

14. Perform the calculation and apply significant-digit/figure rules: $2.51 \times 0.020 / 1.8890$

a) 0.03 b) 0.027 c) 0.0266 d) 0.0270

15. How many significant digits are in **0.300150**?

a) 4 b) 5 c) 6 d) 7

16. Which statement about protons is FALSE?

a) They have a mass of about 1 amu b) They are found in the nucleus
c) They determine the atomic number d) They have a negative charge

17. Which statement is TRUE about isotopes:

a) determined only by proton number b) number 7 in $^{7}_{3}\mathrm{Li}$ shows isotope for this element
c) have a different number of photons d) identical to atomic number *Z*

18. Convert 273 K to °C (nearest integer).

a) -273 °C b) 0 °C c) 273 °C d) 373 °C

19. Electron have a quantum number (*l*) that can have one of four values (0, 1, 2, or 3), but these are represented by letters *s*, *p*, *d* or *f*. Which of these types has ***three*** possible orbitals that have a "dumbbell" or bilobed shape?

a) ***s*** b) ***p*** c) ***d*** d) ***f***

20. Which of these properties of a wave describe the crest-to-crest or trough-to-trough distance of a wave?

a) wavelength b) amplitude c) frequency d) wave speed

21. Which of the following **units** below would be appropriate to refer to the ***mass*** of any substance/matter?

a) grams (g) b) milliliters (mL) c) centimeters (cm) d) meters (m)

22. An element has an **atomic number *Z*** = 23 and a **mass number *A*** = 48. How many protons are in the nucleus of its atom?

a) 23 b) 25 c) 48 d) 71

23. The energy levels of electrons in orbit around nucleus are quantized. What does this mean?

a) Electrons can move between energy levels only with specific amounts of energy
b) Electrons can move between levels using any amount of energy ("continuous" change)
c) Electrons can change their charge when moving between levels
d) Electrons convert from matter into pure energy when changing levels

24. An element has two naturally occurring isotopes with equal relative abundances (50.00% each). One isotope has a mass of 50.00 amu and the other has a mass of 54.00 amu. What is the atomic mass (being a weighted average mass) of this element as it would appear on the Periodic Table??

a) 50.00 amu b) 50.54 amu c) 52.00 amu d) 104.00 amu

25. According to the Aufbau Principle (***n*** + ***l*** rule), which orbital starts filling with electrons immediately after the **2*s*** orbital?

a) **1*s*** b) **1*p*** c) **2*d*** d) **2*p***

Continue to next page for True/False questions

The following statements are either True or False. Select **choice A** for **true**, or **choice B** for **false**

26. Conversion factors like **1 in = 2.54 cm** do not limit the precision (affect the significant digits) of the final calculation.
27. In the scientific method, a tentative explanation for a set of observations is a **hypothesis.**
28. When an electron goes from the higher energy ***n = 2*** shell to the lower energy ***n = 1*** shell in the atom, it is expected to ***emit*** an electromagnetic wave (photon).
29. The mass number (A) of any element in the Periodic Table gives the identity of the element and its proton count
30. Protons and neutrons have about the same mass while electrons are more than a thousand times less massive.
31. ***All*** of the ***zero digits*** in the number **0.01304500** are considered significant.
32. The number **11.34×10^3** is valid scientific notation.
33. An electron that goes from ***n* = 1** shell to ***n* = 3** shell goes from a ground state to an excited state
34. **Density** is calculated as the volume taken up by the matter ***divided by*** mass of the matter.
35. Each and every **orbital** in an atom can hold 1 or 2 electrons, that is a maximum of 2 electrons

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